Prajwal Prajwal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.61
Gurukula pu college,Karadyal	Class XII (Karnataka school examination and assessment board)	2024	93.50%
Helen's english medium high school	Class X (Karnataka secondary education examination board)	2022	97.28%

SKILLS

Programming Languages: C, C++, Python, Verilog and MySQL

Libraries & Frameworks: NumPy, Matplotlib, Scikit-learn, PyTorch, MNE

Tools & Utilities: Git, GitHub, VS Code, Jupyter Notebook, KiCad, LTSpice, LaTeX

Operating Systems: macOS, Linux, Windows

Hardware & Electronics: PCB Design, Arduino, Oscilloscope, Digital Circuit Design, FSM Design, DDR3 Memory Interface

PROJECTS

AI Development Board for Edge OCR | Team Project

KiCad | Zynq-7000 | DDR3 | USB-C | HDMI

- Designed hardware architecture for an **embedded AI/vision system** on a Xilinx Zynq SoC, integrating processor, DDR3 memory, and peripheral interfaces (USB-C, HDMI) for **connected-device applications**
- Performed component selection and interconnect design across compute, memory, and I/O subsystems, building foundational knowledge of **IoT-connected device architecture**
- Investigated high-speed digital design constraints (DDR3 signal integrity, multilayer routing) relevant to real-world embedded product development
- Collaborated within a multi-member team to integrate hardware design with the system's intended **vision/AI processing pipeline**

EEG Signal Processing for Motor Imagery Classification

Python | PyTorch | NumPy | Scikit-learn | MNE | FastAPI | Docker

- Built an end-to-end **AI/ML pipeline** for cross-subject **EEG signal classification**, covering data preprocessing, model training, evaluation, and inference on the **PhysioNet EEG Motor Movement dataset**
- **Applied signal preprocessing techniques including bandpass/notch filtering** and event-based epoching to convert raw multi-channel EEG into model-ready feature representations
- Implemented and benchmarked classical ML (**Band Power + LDA, CSP + LDA**) against deep learning architectures (**EEGNet, Shallow ConvNet, Deep ConvNet**), improving accuracy from ~50% to **72.2%**
- **Validated model generalization through a transfer learning experiment**, achieving **63.3% accuracy** on unseen real-movement data using a model trained on imagined-movement signals
- Developed and containerized a **FastAPI inference service with Docker**, exposing REST endpoints for real-time model inference and end-to-end deployment testing

Digital Signal Processing Algorithms and Signal Analysis

Python | NumPy | Matplotlib

- Implemented core **Digital Signal Processing (DSP)** algorithms from first principles, including **sampling**, **quantization**, **discrete-time systems**, **convolution**, and **Discrete Fourier Transform (DFT/IDFT)**.
- Analyzed and validated **system classification**, **linearity**, **time invariance**, **causality**, **stability**, **sampling theorem**, and **frequency-domain** properties through computational experiments.
- Processed **speech**, **music**, **image**, and **sinusoidal signals** to investigate **signal reconstruction**, **frequency analysis**, and practical DSP applications.

House Price Prediction Using NumPy

Python | NumPy

- Built a linear regression model from scratch using only **NumPy**, implementing feature standardization, batch gradient descent, and model evaluation on a 414-sample real-estate dataset.
- Reduced training MSE from **1635.55** to **79.86** over **1,000** gradient descent iterations, achieving a **test MSE of 67.40** and **MAE of 6.16**.
- Implemented both **gradient descent** and the **closed-form normal equation (Moore-Penrose pseudoinverse)**, comparing optimization approaches and validating prediction performance against baseline models.

RELEVANT COURSES

Core EE: Digital Systems, DSP, Analog Electronics, Control Systems, Semiconductor Devices, Machine Learning

Computer Science: Programming, Data Structures

Math Courses: Matrix Theory, Probability & Random Processes, Differential Equations

EXTRACURRICULAR

Teaching Assistant - Designed, evaluated, and graded coursework and final assignment with helping students and taking relevant session.

Volleyball - Actively participate in college-level volleyball tournaments and inter-hostel competitions.